QP CODE: 25901635

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026
SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCCMA101

RESPONSIVE WEB DESIGNING
(2024 ADMISSION ONWARDS)

SCHEME FOR EVALUATION

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A

Very Short Answer Questions. Answer all questions . Each question carries **2** marks

1. Define function calling.
   Function calling is the process of invoking a function by using its name followed by parentheses, causing the function code to execute.
2. Differentiate between .focus() and .blur() methods ?
   .focus() gives focus to an input element.
   .blur() removes focus from an element.
3. Write a code snippet to create a responsive navigation bar in Bootstrap?
   Any valid Bootstrap navbar code. Mention responsive classes such as navbar. Give marks.
4. What is one thing that all popular web browsers have in common when it comes to JavaScript?
   All modern browsers contain a built-in JavaScript engine and support JavaScript execution. Examples of javascript engine V8,spidermonkey, nitro etc.(Value liberally)
5. Identify the purpose of .container-fluid class.
   Creates a full-width container that spans the entire viewport width .
   The main uses are- Website headers and footers, Full-width images ,Admin dashboards, Responsive layouts etc.
   Mention any one give marks.

PART B

Short Answer Questions. Answer any 5 questions. Each carries 5 marks.

6. Explain the difference between a variable's declaration and initialization
   Declaration: creating a variable name. eg: let x;
   Initialization: assigning an initial value. Eg: x=10;
   Declaration with initialisation: Eg: let x=10;

7. Discuss the real-time feedback in the context of form validation.

   Real-time feedback in form validation means checking the user's input **immediately while they are typing or as soon as they leave a field**, instead of waiting until the entire form is submitted. When a user enters data into a form field, JavaScript validates the input instantly and displays feedback such as:Valid input ,Error messages ,Warnings or suggestions etc.

   Definition with example 5 marks. ,Definition only 3 marks , Example only 4 marks .

8. Identify the class in Bootstrap which is used to make a webpage's design responsive.

   In Bootstrap, the primary class used to create a responsive layout is the **.container** class (or **.container-fluid** for full-width layouts).

   Bootstrap's responsiveness is mainly powered by grid classes such as:

   .row , .col- , .col-sm-* , .col-md-* ,.col-lg-* , .col-xl-*

   Give marks for explanation and examples about container classes or grid classes.

9. Explain the statement: "JavaScript helps in validating a form before it is submitted."

   .JavaScript can check the data entered by a user **before the form is sent to the server**. This process is called **client-side form validation**. Form validation is the process of verifying that user input is: Complete , Correctly formatted ,Within acceptable limits

   Explain the form validation with an example code.

10. Explain any two basic filters in jQuery with examples.

    :first, :last, :even, :odd, :eq(),:contains()

    Explain any two with example code

11. Discuss various text color classes in Bootstrap

    In Bootstrap, text color classes are utility classes used to quickly change the color of text without writing custom CSS. These classes begin with the prefix text-.

    Examples :

    text-primary, text-secondary, text-success, text-danger, text-warning, text-info, text-light, text-dark, text-muted.

    Give an example code with any 3 classes .

12. Describe the working of switch statement with syntax or proper example.

    The switch statement in JavaScript is used to select one block of code from many possible options based on the value of an expression.

    Syntax:

```
switch (expression) {
   case value1:
      // Code to execute if expression === value1
      break;
   case value2:
      // Code to execute if expression === value2
      break;
   case value3:
      // Code to execute if expression === value3
      break;
   default:
      // Code to execute if no case matches
}
```

    Explain with example.

PART C

Essay Questions.Answer any 1 question. Each question carries 15 marks .

13. Describe how jQuery allows you to add new DOM elements. Provide examples of methods such as .append(), .prepend(), .after() and .before() ?

    In jQuery, new HTML elements can be inserted dynamically into the Document Object Model (DOM). This allows webpages to update content without reloading.

    (give 2 marks)

    A new element can be created using jQuery:

    ```
    let newItem = $("<p>This is a new paragraph</p>");
    ```

    This creates a paragraph element but does not insert it into the page until an insertion method is used.(give 3 marks)

    Explain the methods such as .append(), .prepend(), .after() and .before() with their purpose and examples. Give 2.5 marks for each method

14. Explain the concept of responsive web design and how Bootstrap helps to achieve it. Provide examples.

    Definition: Responsive Web Design is an approach where a website automatically adjusts its layout, images, and content to fit different screen sizes such as:

    - Mobile phones
    - Tablets
    - Laptops
    - Large desktop screens

The main goal is to ensure the website looks good and is usable on all devices without creating separate versions for each device.(Give 3 Marks)

Key Principles of Responsive Design

1. Fluid Grid Layout:Uses flexible grids instead of fixed-width layouts so elements resize based on screen width.
2. Flexible Images:Images scale automatically within their containers.
3. Media Queries: CSS rules that apply different styles depending on screen size.(Give 3 Marks)

Expalin how Bootstrap Helps in Responsive Web Design

In Bootstrap, responsiveness is built-in using a combination of:

1. Grid System: Bootstrap uses a 12-column grid system that automatically adjusts based on screen size.
2. Responsive Breakpoints: Bootstrap defines screen sizes such as:
    - sm (small devices)
    - md (medium devices)
    - lg (large devices)
    - xl (extra large devices)
3. Responsive Utility Classes: Bootstrap provides classes to control layout and visibility easily.

Explain each with examples. (Give 3 marks each)

Prepared by,

Seema K S, Assistant Professor, TOMS Arts and Scince College, Mattakara

9495444211